HOW IT WORKS

How a print job runs a language model

TONER-1 is the most capable language model ever to run entirely inside a print job.* It is one 595 KB PostScript file. Send it to a laser printer and the printer's own processor writes a short story, usually about a printer, then prints it. The computer that sent it does no math.

01

A print job is a program

PostScript, the language most laser printers speak, is not a file format but a complete programming language, stack-based like Forth. Send `3 4 add` and the interpreter leaves 7 on its stack. It has variables, loops and floating-point math. A print job is simply a program; the printed page is a side effect of the drawing commands it calls.

02

Printers are computers

To run those programs, every PostScript printer carries its own processor and memory. The 1985 Apple LaserWriter had a 12 MHz Motorola 68000 and 1.5 MB of RAM, a faster CPU than the original Macintosh it plugged into (a 68000 at about 7.8 MHz). Modern printers have far more, and spend most of it waiting for spreadsheets.

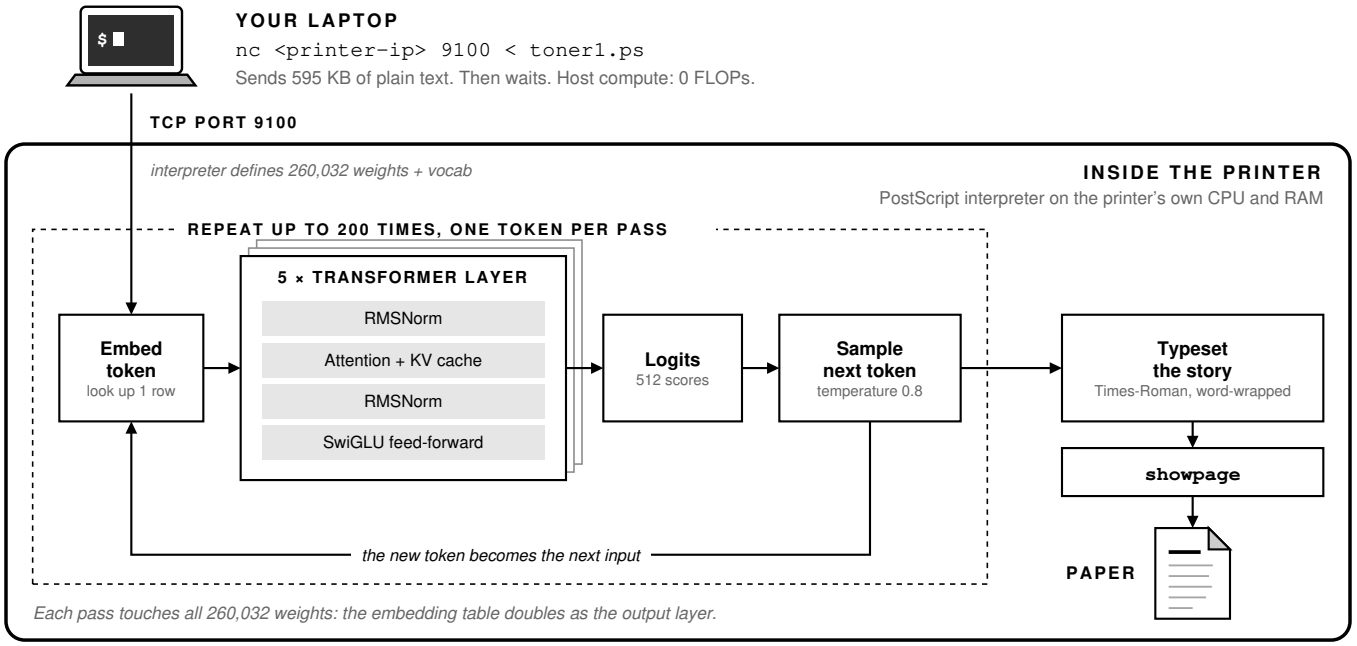
03

An LLM is just arithmetic

Strip away the mystique and a language model is a long list of multiplications and additions: matrix-vector products, a normalization step, a softmax that turns scores into probabilities, and a weighted coin flip that picks the next word piece. Any language with loops and multiplication can run one. PostScript has both.

THE PIPELINE

What happens after you send toner1.ps to port 9100



260,032
parameters
int8, one scale per row

595 KB
the entire print job
weights + vocab + program

≈80 tok/s
in Ghostscript 10.07
Apple-silicon Mac, measured

0.3–5 tok/s
on a real printer
estimated, not yet measured

0 FLOPs
on the host
the laptop only sends bytes

** Among int8, 260,032-parameter, English-language, children's-story models executed by a PostScript interpreter, as of this printing. Ghostscript counts as a printer for the purposes of this footnote. Model: Karpathy's llama2.c stories260K, trained on TinyStories, then fine-tuned by us on 8,757 printer stories written by Xing4.0-29B-A4B (Apache-2.0). Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.*

INSIDE THE FILE

What's inside the 595 KB print job

BYTES IN TONER1.PS BY PURPOSE: 595,372 TOTAL, 455 LINES OF PLAIN TEXT



THE DATA, LINES 1 TO 106

```
 1 %!PS-Adobe-3.0
 7 /D 64 def /HID 172 def /NL 5 def /NH 8 def /NKV 4 def
     /V 512 def /SEQ 512 def /HS 8 def /KVD 32 def
     /KVMUL 2 def /NPARAMS 260032 def
```

```
16 /WQ [
17 [[0.00261005 0.00431596 0.00340043 0.00252913 ... 0.0012534]
18 [<96917e84906c8a8537147183bfa939875d25794a...> <ae99...> ...]]
```

```
row 0 bytes        96       91       7e       84
minus 128          22       17       -2        4
× 0.00261005  0.0574   0.0444  -0.0052   0.0104
```

1 **The model card is one line.** Line 7 defines width 64, 5 layers, 8 query heads sharing 4 key/value heads and a 512-token vocabulary. Nothing runs yet; names are just defined. Line 1 makes it a print job; lines 2 to 6 are comments for the spooler.

2 **Weights are bytes.** For each matrix row, export.py takes the largest absolute weight m and stores scale = m/127. Each weight is rounded to an integer from -127 to 127, plus 128, and written as one byte of a hex string. Two hex characters per weight: that is most of the file.

THE PROGRAM, LINES 107 TO 455

```
129 % o x M mv  ;  M = [scales rows], rows ... offset by 128
130 /mv { /mM exch def /mx exch def /mo exch def
131   /mS mM 0 get def /mQ mM 1 get def /mn mx length 1 sub def
132   /msx 0 mx { add } forall 128 mul def
133   0 1 mQ length 1 sub { /mi exch def /mrow mQ mi get def
134     0 0 1 mn { dup mrow exch get exch mx exch get mul add } for
135     msx sub mS mi get mul mo mi 3 -1 roll put } for
136 } bind def
```

3 **mv does the heavy lifting.** Matrix times vector: about 259,000 multiply-adds per token run through these eight lines. `get` on a string returns a byte as an integer. Rather than subtract 128 from every byte, mv takes the raw dot product, subtracts `msx` (128 times the sum of the input), then applies the row's scale.

```
139 /rope { /rn exch def /rv exch def
140   0 2 rn 1 sub { /ri exch def
141     /hd ri HS mod def
142     pos 10000 hd HS div exp div 57.29577951 mul /ang exch def
143     /fcr ang cos def /fci ang sin def
```

4 **RoPE, in degrees.** Rotary embeddings turn query/key pairs by `pos / 10000^(hd/8)` radians, but PostScript's `sin` and `cos` take degrees. Hence 57.29577951 (180/pi). Without it, every position would get the wrong rotation.

```
118 SEED srand 4 { rand pop } repeat
204 /r rand 2147483647 div ssum mul def /cum 0 def
      /pick V 1 sub def
205 0 1 V 1 sub { /j exch def /cum cum logits j get add def
      cum r gt { /pick j def exit } if } for
```

5 **Sampling with rand.** The logits become a softmax at temperature 0.8. PostScript's built-in `rand` picks a point along the total, and the loop walks the running sum until it passes it. Line 118 skips four tiny first draws. Seed 42: same interpreter, same story.

```
257 % ... typesetting: the same print job lays out the page ...
368 fs /Times-Roman F ...
449 9 /Helvetica-Bold F 72 79 moveto (Host compute: ...) show
453 showpage
```

6 **Then, and only then, toner.** After 200 tokens (or an end-of-story token), the same job word-wraps the story in Times-Roman, draws the run card and calls `showpage`. Until then, the printer was just a computer.

PRINT IT YOURSELF — BRING YOUR OWN PRINTER

```
nc <printer-ip> 9100 < toner1.ps
```

TONER-1 Pro: toner1_pro.ps, 12.2M parameters, 24.9 MB, ours. Same command. About 28 times slower.

- **Needs real PostScript.** A PostScript interpreter, Level 2 or later, is required. PCL- or GDI-only printers may print the file as raw text.
- **Mind the limits.** Long jobs can hit the job timeout or the VM limit. Ghostscript uses about 3.5 MB of VM.
- **Expect minutes.** An estimated 0.3 to 5 tok/s is roughly 1 to 10 minutes per 200-token page. It is thinking.
- **Ask first.** Only use printers you own or have permission to use. The queue behind you has feelings.

STATUS ☑ Verified in Ghostscript 10.07: 200 tokens, about 80 tok/s, about 3.5 MB of VM ☐ **Not yet run on a physical printer**

Excerpts are verbatim from toner1.ps; gray . . . marks trims, and long lines are re-wrapped or dedented. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in Ghostscript only; PSChess played chess on real printers. TONER-1 is built to run on an actual printer; that run is pending. Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.

TONER-1: a frontier model with a paper trail.

State of the art among language models that run entirely inside a print job.

> The most capable open-weight, US-Letter-native, sub-300K-parameter, PostScript-native, zero-host-FLOP, duplex-ready language model running on hardware already in your copy room. *Evaluated on 20 lb bond at room temperature.*

SPECIFICATIONS

Base model	Karpathy's llama2.c stories260K (trained on TinyStories), fine-tuned by us on 8,757 printer stories written by Xing4.0-29B-A4B (Apache-2.0)
Parameters	260,032
Architecture	Llama-2 decoder: 5 layers, dim 64, 8 query heads / 4 KV heads, head size 8, SwiGLU FFN (hidden 172), RoPE, RMSNorm
Vocabulary	512-token SentencePiece
Quantization	int8 weights, one scale per row; activations in PostScript reals; embedding shared with the output head
Context window	8.5 × 11 in. (512 tokens)
KV cache	Persistent, archival, recyclable
Job size	595 KB, one file: `toner1.ps`. Weights, vocabulary, a forward pass of about 100 lines and its own page layout
Speed, measured	About 80 tok/s in Ghostscript 10.07 on an Apple-silicon Mac, in about 3.5 MB of PostScript VM
Speed, printer	0.3 to 5 tok/s, estimated, not yet measured. A 200-token page takes roughly 1 to 10 minutes
Time to first token	Includes fuser warm-up
Accelerator	The office's hottest inference chip (fuser, typically about 200 °C)
Host compute	0 FLOPs. The host only sends bytes: `nc <printer-ip> 9100 < toner1.ps`
Output modality	Native toner
Open weights	Literally. Hex appendix available; OCR to load
Pricing	Inference cost includes toner

INTENDED USE

Writing short children's stories, mostly about printers, one US Letter sheet at a time, on a printer you own or have permission to use. Demonstrating that a print job is a program. *Out of scope: prompts. The model chooses the plot.*

LIMITATIONS

- It is 260K parameters.
- It mostly writes about Lily, Tim and the printer.
- It will hallucinate a quiet salad.
- No prompt input: every story starts from the beginning-of-sequence token.
- Needs a PostScript interpreter, Level 2 or later. PCL- or GDI-only printers may print the file as text.
- Long jobs may hit the printer's job timeout or run out of VM.

SAFETY

- Refuses to generate while Tray 2 is open. Close the tray to proceed.
- **Air-gapped mode:** unplug Ethernet once the job has uploaded.
- **Kill switch:** the power button.
- Outputs are fused to paper and can no longer execute.

PRODUCT LINEUP

TONER-1 — AVAILABLE

260K, PRINTER-TUNED / 595 KB JOB

The everyday copy-room model, now fine-tuned on printer stories. Under 4 s per page in Ghostscript; an estimated 1 to 10 minutes on a printer.

TONER-1 Pro — EST. 20 MIN–5 H

12.2M, OURS / 24.9 MB JOB

Trained from scratch by us, 47 times the parameters. Estimated: send it after lunch, collect your story by close of business.

TONER-1 Classic — PLANNED

1985 LASERWRITER / 68000

A 1985-native AI PC. The LaserWriter's 12 MHz 68000 was faster than the 7.8 MHz Macintosh it was plugged into.

TONER-1 is a working print job and a parody of frontier-model launches. This handout is a PostScript program, too.
Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.

Undefeated inside a print job.

TONER-1 posts a result on every benchmark it can enter. Frontier models from major labs were offered identical conditions: one print job, one sheet, zero host FLOPs.

FIGURE 1 HEADLINE RESULTS (PARODY)

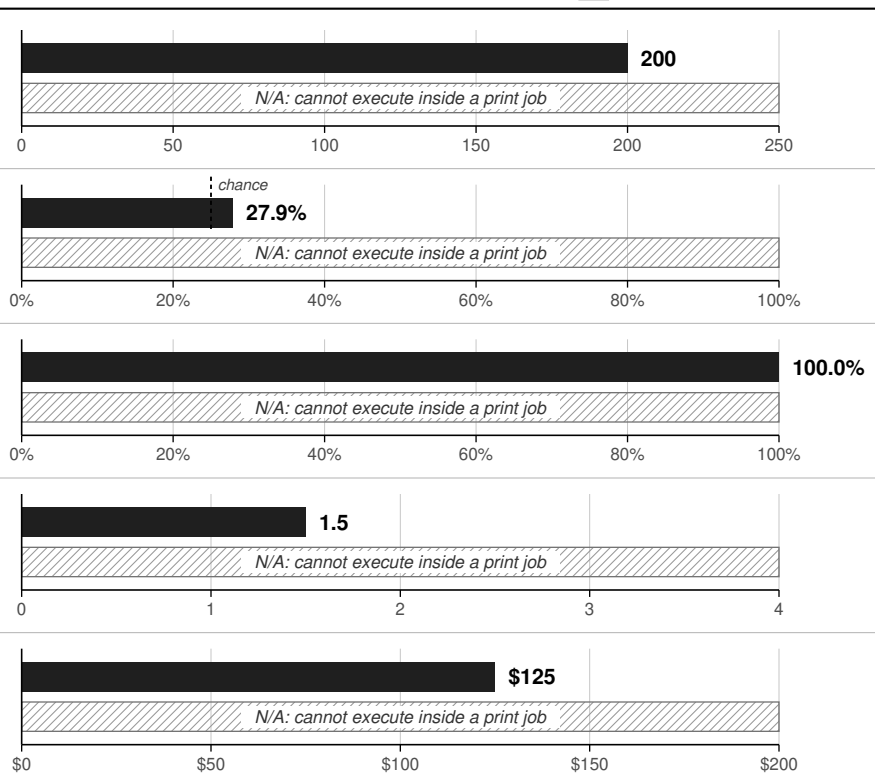
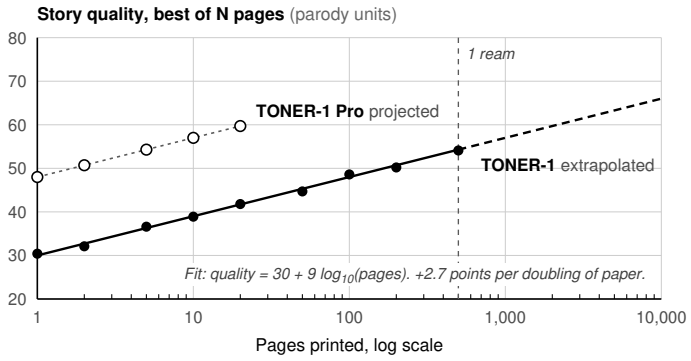


FIGURE 2 TEST-TIME SCALING



EVALUATION SETUP

Interpreter	Ghostscript 10.07, Apple-silicon Mac
Printer	Office laser, PostScript emulation (pending)
Speed	Measured in Ghostscript; printer estimated
Paper	20 lb bond, US Letter, simplex
Sampling	Temperature 0.8, seed 42, 200 steps
Prompt	None. Every run starts from BOS.
Ambient	Room temp.; fuser typically about 200 °C
Frontier models	Not run: no print-job support

REPRODUCE

```
nc <printer-ip> 9100 < toner1.ps
```

Parody benchmarks. Not real evaluations. TONER-1 speed measured in Ghostscript; printer speed estimated. Frontier models from major labs were not run on these benchmarks and are not named; N/A means only that they cannot execute inside a print job. Figure 2 is parody data. TONER-1 is Karpathy's llama2.c stories260K, fine-tuned by us on printer stories, exported to int8 and run by a PostScript forward pass. TONER-1 Pro is a 12.2M-parameter model we trained from scratch, with our own 4,096-token tokenizer. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in PostScript under Ghostscript, and PSChess played chess on real printers. TONER-1 is built to run on an actual printer; that run is pending. Only use printers you own or have permission to use.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

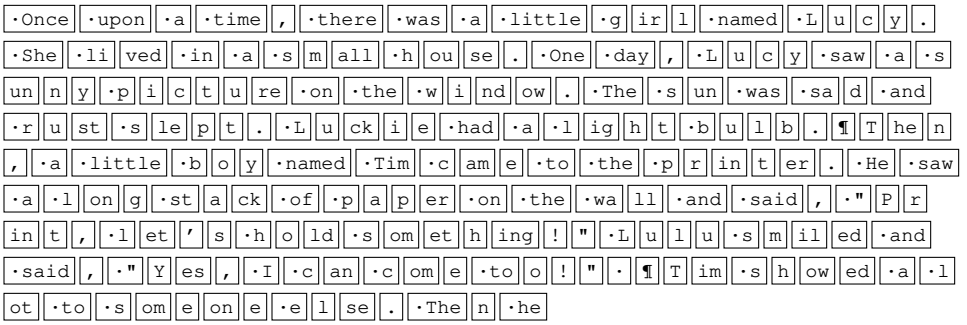
Once upon a time, there was a little girl named Lucy. She lived in a small house. One day, Lucy saw a sunny picture on the window. The sun was sad and rust slept. Luckie had a light bulb.

Then, a little boy named Tim came to the printer. He saw a long stack of paper on the wall and said, “Print, let’s hold something!” Lulu smiled and said, “Yes, I can come too!”

Tim showed a lot to someone else. Then he…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.



HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	2.4	82.3	20	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a lonely drop of ink. The drop was getting shiny and had a soccer ball. He wanted to read the map to the printer. But when he went to a clock, he saw a yellow ball and gave it to his friend. The ball looked at it and wanted to visit him.

The drop of ink showed Rosa the ball and crossed the wrong wing. The printer took it and found the machine inside. The printer was so surprised! It was a…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·there	·was	·a	·l	on	e	l	y	·d	r	o	p	·of	·in	k	.
·The	·d	r	o	p	·was	·g	et	t	ing	·s	h	in	y	·and	·had	·a	·so	c	c	er
·b	all	.	·He	·want	ed	·to	·	re	ad	·the	·m	a	p	·to	·the	·p	r	in	t	er
.	·B	ut	·w	he	n	·he	·w	ent	·to	·a	·c	l	o	ck	,	·he	·saw	·a	·y	e
ll	ow	·b	all	·and	·g	a	ve	·it	·to	·his	·friend	.	·The	·b	all	·l	oo	ked	·a	t
·it	·and	·want	ed	·to	·	v	is	it	·h	im	.	·	¶	T	he	·d	r	o	p	·of
·in	k	·s	h	ow	ed	·	R	o	s	a	·the	·b	all	·and	·c	r	o	s	s	ed
·the	·w	r	on	g	·w	ing	.	·The	·p	r	in	t	er	·to	o	k	·it	·and	·f	ou
nd	·the	·m	a	ch	in	e	·in	s	id	e	.	·The	·p	r	in	t	er	·was	·so	·s
u	r	p	r	is	ed	!	·I	t	·was	·a										

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	2.4	83.0	40	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

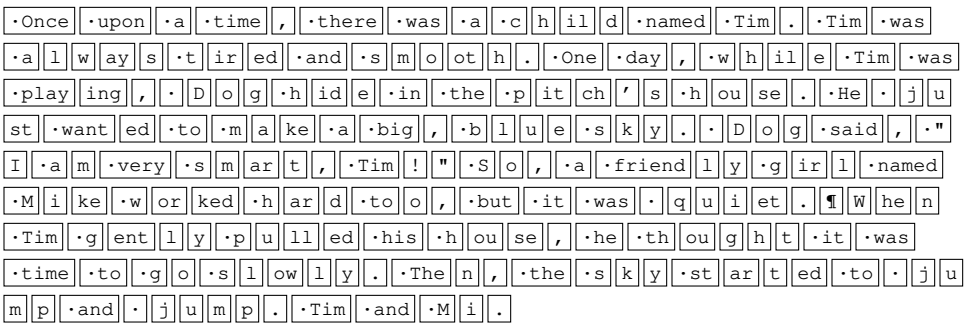
Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a child named Tim. Tim was always tired and smooth. One day, while Tim was playing, Dog hide in the pitch's house. He just wanted to make a big, blue sky. Dog said, "I am very smart, Tim!" So, a friendly girl named Mike worked hard too, but it was quiet.

When Tim gently pulled his house, he thought it was time to go slowly. Then, the sky started to jump and jump. Tim and Mi.

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.



HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	2.4	83.0	22	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200, reached before the end-of-story token.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

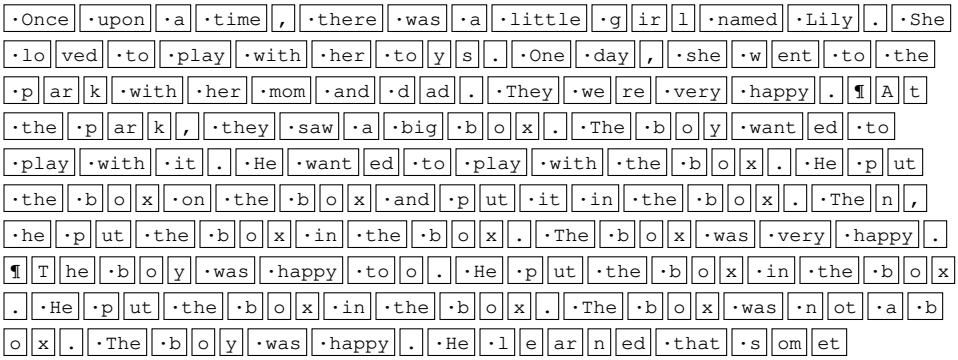
Once upon a time, there was a little girl named Lily. She loved to play with her toys. One day, she went to the park with her mom and dad. They were very happy.

At the park, they saw a big box. The boy wanted to play with it. He wanted to play with the box. He put the box on the box and put it in the box. Then, he put the box in the box. The box was very happy.

The boy was happy too. He put the box in the box. He put the box in the box. The box was not a box. The boy was happy. He learned that somet…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.



HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**2.4**	**82.0**	**42** (unused)	**0** greedy

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Lily came to the park with sand a large whirt bepying balls. Lily has yellow laes that were big and empty. “Quack shell are new!” said Lily.

“Meer, mom, it is on the side,” Anne and nodked. “Hach wind been theumse it hednd!”

Lily washed the last again. The lazy cat was that moving. Lily purred and Miri. They were like something for dienen, playing with each faucet note but from her new one. Lily’s dif…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Lily	·c	am	e	·to	·the	·p	ar	k	·with	·sa	nd	·a	·l	ar	g	e	·w	h	ir	t	·be
p	y	ing	·b	all	s	.	·Lily	·ha	s	·y	e	ll	ow	·l	a	es	·that	·we	re	·big	·and
·e	m	p	t	y	.	·"	Q	u	a	ck	·she	ll	·a	re	·ne	w	!	"	·said	·Lily	.	¶	"	M	e
er	,	·mom	,	·it	·	is	·on	·the	·s	id	e	,	"	·	A	n	n	e	·and	·n	o	d	ked	.
·"	H	a	ch	·w	i	nd	·be	en	·the	u	m	se	·it	·he	d	nd	!	"	¶	L	ily	·was	he	d
·the	·l	a	st	·a	g	a	in	.	·The	·l	a	z	y	·c	at	·was	·that	·m	o	v	ing	.		
·Lily	·p	u	r	r	ed	·and	·M	ir	i	.	·They	·we	re	·li	ke	·s	om	et	h	ing	·for			
·d	i	en	en	,	·play	ing	·with	·e	a	ch	·f	a	u	c	et	·n	ot	e	·but	·f	r	om			
·her	·ne	w	·on	e	.	·Lily	’	s	·d	i	f														

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	2.4	82.6	6	1.4

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1 Pro

12,194,688 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, a little bear sat on a table near a printer. The printer had a red one. The bear wanted a new friend, but he did not have a picture of a cake. The printer was sad. The printer beeped softly and blinked. “I am four years old,” the printer said. “Do you want me to eat you?” the teddy bear nodded. The printer took a bite of honey. It made a funny face. The bear laughed. “That is my cake!” he said. Then the printer heard a loud noise. It was the printer. The printer was printing a big cake. The bear wanted to eat the cake. The printer stopped beeping. The printer beeped once. “I am sorry,” the printer said. The bear came to help. The bear threw another piece of cake on the floor. The printer ate another piece. Each page was good at printing. The printer hummed and printed the cake. Everyone was happy. The printer learned to…

TOKEN STREAM

The first 70 of 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·a	·little	·bear	·sat	·on	·a	·table	·near	·a
·printer	.	·The	·printer	·had	·a	·red	·one	.	·The	·bear	·wanted	·a	·new
·friend	,	·but	·he	·did	·not	·have	·a	·picture	·of	·a	·cake	.	·The
·printer	·was	·sad	.	·The	·printer	·beeped	·softly	·and	·blinked	.	·"	I	
·am	·four	·years	·old	,"	·the	·printer	·said	.	·"	Do	·you	·want	·me	·to

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**71.1**	**2.81**	**42**	**0.8**

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	TONER-1 Pro 12M (Llama-2 architecture) · dim 384 · 6 layers · 6 query / 6 KV heads · 4,096-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.